COMP3033/COMP9033
Cloud and Distributed Computing

Big Data and Cloud Services in Healthcare

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Objectives

- Understand big data and its characteristics
- Learn about big data technologies in healthcare
- Explain how cloud computing can address big data challenges in healthcare

Big Data

Big Data

What is Data?

In computing, data is information that has been translated into a form that is efficient for storage and/or processing.

- Every day, we create 2.5 quintillion bytes of data.
- 500 million tweets sent every day
- 5.75 billion Facebook likes every day
- 4.3 billion Facebook messages posted every day
- 6 billion daily Google searches!

What is Metadata?

We are **Tracing** everything:

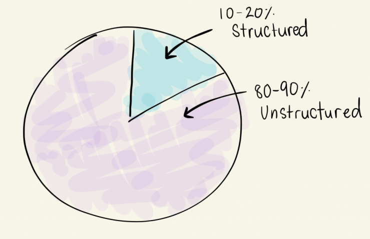
- What is happening?
- Who is doing it?
- Where is it happening?
- When?
- Why?
- How?

For example, smart phones track

- Our location
- Our health data
- Our speed
- What apps we are using
- What musics we listen to

Data Type

- Data can be classified as
  - Unstructured
  - Structured
- Majority of data being created is unstructured

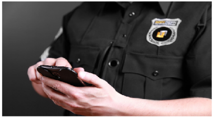


Big Data?

Police Investigation

IoT Data

Private Data



What is Big Data?

- Defined as large set of structured, unstructured and disorganised data.
- Refers to our ability to collect and analyse the ever expanding amounts of **data** and **meta-data** that we are generating every second
- Can be seen as a massive number of small data islands
- Examples
  - Traditional enterprise data
  - Machine-generated data
  - Sensor data
  - Social data

Big Data Characteristics

Big Data Characteristics

Characteristics of Big Data – 7 Vs

- Volume
  - The volume or size of data being created.
  - Big data is drawn from many sources, resulting in huge amounts of disorganised data that cannot be interacted with in traditional ways, such as database queries that rely on normalised data sets.
- Velocity
  - The speed at which data is gathered. Timely processing of incoming data is important to generate useful information when it is needed.
    - Storage of arriving data, as the capacity to process it is lower than the speed at which it arrives
    - Forcing a response to data as it arrives, in order to act on time-sensitive data.

Characteristics of Big Data – 7 Vs (Cont.)

- Variety
  - Big data contains a huge variety of data, from images, video, audio, text, and other forms.
  - This disorganised variety of data makes it very difficult to establish a system that properly integrates data.
- Veracity
  - This point refers to the truthfulness of the data; i.e. whether the data is of the type it claims to be, and whether it is clean and precise.
  - Tools and algorithms must be used to clean the data and preserve the integrity of the database.

Characteristics of Big Data – 7 Vs (Cont.)

- Validity
  - Validity is concerned with the correctness and accuracy of data, specifically in regard to its use.
  - Data may be verified, but may not be relevant or applicable for the use in question.
- Value
  - A ‘special’ point.
  - Value doesn’t refer to the analysis or storage of big data, rather it refers to the desired outcome of big data processing.
- Visibility
  - Irrespective of the format, data should be easily readable, understandable, and accessible.

Big Data Storage

- Traditional relational databases are not effective for the storage of big data
- NoSQL Database ✓
- Examples of NoSQL Database
  - MongoDB
  - Redis
  - Cassandra
  - CouchDB
  - HBase

Importance of Big Data

- When big data is refined it can
  - Enhance productivity
  - Provide a stronger competitive position
  - Improve innovation
- To make use of the big data it has to be
  - Acquired
  - Organised
  - Analysed

Big Data Platform

Big Data Acquisition

- Need infrastructure for streams of high velocity and high variety.
- NoSQL database are frequently used.

Big Data Organisation

- Data integration
- Need infrastructure to process and manipulate data
- Need database to allow large data volumes to be organised and processed
- Need programs to aggregate results

Big Data Analysis

- Completed in distributed environment
- Need infrastructure for statistical analysis and data mining for diverse data

Sectors That Use Big Data

- Healthcare services
- Manufacturing companies
- Social media sites such as Facebook and LinkedIn
- Retailers who use context aware computing
- Transportation services
- Utility companies
- ...

Big Data and Cloud

Big Data and Cloud

Big Data and Cloud Computing

- One of the vital issues that organisations face with the storage and management of Big Data is the huge amount of investment to get the required hardware setup and software packages.
- Some of these resources may be over-utilised or under-utilised with varying requirements over time. We can overcome these challenges by providing a set of computing resources that can be shared through cloud computing.
- The cloud computing environment saves costs related to infrastructure in an organisation by providing a framework that can be optimised and expanded horizontally.

In-Memory Technology for Big Data

- Hardware obstructions and limitations, lag of memory indifferences have to be side-lined and streamlined with something faster like cache memory or dynamic access memory so that the data is readily available for disposal.
- The in-memory big data computing tool supports the processing of high-velocity data in real-time and also the faster processing of stationary data.
- Technologies like event-streaming platforms, in-memory databases and analytics, and high-level messaging structures are witnessing massive growth that resonates with the organisational needs for better understandings achievable by a wider and deeper data assessment.

Big Data Techniques

- To analyse the datasets, there are many techniques available, some of which are as follows:
  - Massive Parallelism
  - Data Distribution
  - High-Performance Computing
  - Task and Thread Management
  - Data Mining and Analytics
  - Machine Learning
  - Data Retrieval

Massive Parallelism

- According to the simplest definition available, a parallel system is a system where multiple processors are involved and associated to carry out the concurrent computations.
- Massive parallelism refers to a parallel system where multiple systems interconnected with each other pose as a single mighty conjoint processor and carry out tasks received from the data sets parallelly.
- To meet the Big Data dynamics, systems do scale up their operational efficiency using a massive parallelism system that can comfortably work with extremely large datasets parallelly.

Data Distribution

- There are various approaches to data distribution in a Big Data system, described as follows:
  - **Centralised Approach**: A central repository is used to store and download the essential dataset by virtual machines.
  - **Semi-Centralised Approach**: Semi-centralised approach reduces the stress on the networking infrastructure.
  - **Hierarchical Approach**: In a hierarchical approach, the data is fetched from the parent node, i.e., the virtual machine, in the hierarchy.
  - **P2P Approach**: P2P streaming connections are based on hierarchical multi-trees.

High-Performance Computing

- High-performance computing is the simultaneous use of supercomputers and parallel processing techniques for solving intricate computation problems.
- It emphasises making parallel processing systems and algorithms by joining both parallel and administrative computational methods.
- The words supercomputing and high-performance computing are often used to resemble each other.
- High-performance computing is used for performing research activities and cracking advanced problems through computer simulation, modelling and analysis.

Task and Thread Management

- **Threads** are simply the OS-based feature with their own Kernel and memory resources, and allow application logic to be segregated into concurrent multiple execution paths. It is a useful feature when complex applications having multiple tasks need to be performed at the same time.
- **Task parallelism** refers to the execution of computer programmes throughout the multiple processors on the different or same machines. It emphasises on performing diverse operations in parallel to best utilise the accessible computing resources like memory and processors.
- **Data parallelism** focuses on the effective distribution of datasets throughout the multiple calculation programs.

Data Mining and Analytics

- Data mining is the process of data extraction, evaluating it from multiple perspectives and then producing the information summary in a meaningful form that identifies one or more relationships within the dataset.
- Data analysis is an experiential activity, where the data scouring gives out some insight.
- Data analytics is about applying an algorithmic or logical process to derive insights from a given dataset. For example, looking at the past year's weather and pest data, for the current month, we can determine that a particular type of fungus grows often when the humidity levels reach a definite point.

Machine Learning

- Machine learning formally focuses on the performance, theory and properties of learning algorithms and systems. Machine learning is considered an ideal research field for taking advantage of the opportunities available in big data.
- It delivers on the potential of mining the value from huge and different data sources with less dependence on human instructions. It is data-driven and runs at a machine scale and is well-suited to the complication of dealing with different data sources and the enormous range of variables and quantities of data involved.
- Machine learning systems utilise multiple algorithms to discover and show the patterns hidden in the datasets.

Data Retrieval

- Big Data systems retrieve data using a distributed processing framework.
- It helps programmers in solving parallel data problems where the dataset can be divided into small chunks and handled autonomously.

Big Data, Cloud and Healthcare

Big Data, Cloud Computing and Healthcare

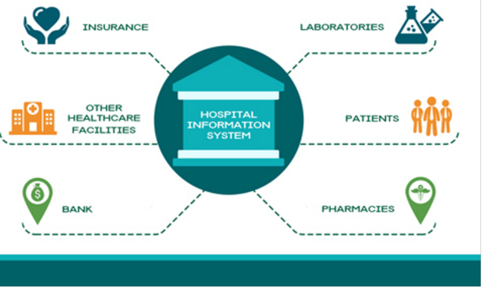
Healthcare

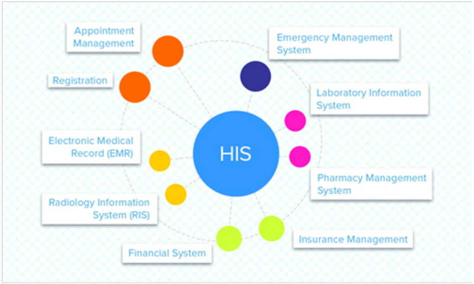
- The act of taking preventative or necessary medical procedures to improve a person's well-being. This may be done with
  - Surgery
  - The administering of medicine
  - Other alterations in a person's lifestyle.
- The services rendered by members of the health professions for the benefit of a patient.
- These services are typically offered through a healthcare system made up of hospitals and physicians.

Hospital Information System (HIS)

- A HIS is essentially a computer system that can manage all the information to allow healthcare providers to do their jobs effectively.
- These systems have been around since they were first introduced in 1960s and have evolved with time and the modernisation of healthcare facilities.
- Modern HIS includes many applications addressing the needs of various departments in a hospital.
  - They manage the data related to clinic, finance department, laboratory, nursing, pharmacy, radiology, and pathology departments.
- Web based HIS are often used between two or more hospitals at different locations to share relevant data and information.
- However, there is a need for cloud-based systems due to the current market dynamics in healthcare.

Hospital Information System





Source: `http://blog.azoft.com/hospital-information-system`

Current Market Dynamics in Healthcare

- There is a substantial growth for healthcare services due to
  - Aging population
  - Increasing prevalence of chronic diseases
  - Increasing interest in wellness by consumers
- Concurrently, there are cost pressures stemming from the need to do more and high-quality work with fewer and more costly resources and reduced revenue.
- Continual change in both supply and demand within the healthcare market influences the use of IT and serves as the principal driver for the adoption of cloud computing.

Systems Needed by Healthcare

- The need to pass patients' records from organisation to organisation with standardised patient Electronic Health Records (EHRs), requires storage, security, and web standards.
- Doctors need software, tools and computing systems to store and query information
  - For example, on huge MRI (Magnetic Resonance Imaging) and electrocardiogram image databases, yielding diagnoses in minutes.
- healthcare and treatment are integrated using
  - Connected healthcare
  - Big data analytics
  - Telemedicine
  - IoT-enabled healthcare
  - Diagnostic support
  - Backup and disaster recovery services

Connected Healthcare

- A model for healthcare delivery that uses technology to provide healthcare seamlessly across multiple providers.
- Cloud computing can provide new and unique opportunities for patients to engage with medical staff to better manage their care.
- Leverages emerging technologies to enable care outside of hospital or doctor's office, using mobile and wearable devices that connect to cloud-based smart healthcare systems.

Big Data Analytics

- Big Data analytics provides the mechanisms for aggregating, mining, and analysing large amounts of medical data, which improves processes, treatments, effectiveness, costs, and conditions.
- Cloud-based big data solutions like genomic research can securely be shared among authorised healthcare organisations to provide access to potentially life-saving information and accelerate research and development.

Telemedicine

- Provides the opportunity to offer
  - Highly specialised and preventative medical advice to rural areas
  - Remote consultations e.g., treating cataracts before they cause blindness.
- Cloud computing is well suited to providing the connectivity channels required to support telemedicine, avoiding the need to install and support in-house which can be complex and may require specialised technologies.

IoT Enabled Healthcare

- With the plethora of new consumer wearable devices and application specific home health monitoring devices able to provide real-time data, IoT enabled healthcare is becoming more and more appealing for patients of all age ranges.
- These devices can provide vital signs data from a patient anywhere in the world to the hospital staff. It allows them to monitor a patient's health while giving the patient the flexibility to live their life.
- Cloud systems can make it straightforward to handle the connectivity requirements of IoT devices, from registration to support of IoT communication protocols and the management of device data.

Diagnostic Support

- Services to assist in diagnosis are becoming a useful tool to guide medical practitioners, helping to reduce time spent, and improving productivity.
- Cloud-based systems are gaining cognitive features, based on learning from very large datasets of medical data.
- These features allow physicians, nurses, and other healthcare providers to use multiple cognitive solutions that help analysing big data from global sources, correlating against specific data flagged on patients, and assisting in decision-making to service providers.

Backup and Disaster Recovery Service

- An approach to backup data and retention where organisations outsource their backup and run recovery services whenever required.
- In the event of a disaster, the service provider must meet the defined recovery time and recovery point objectives to ensure timely availability of the affected systems with minimal losses of data during an outage.
- The emphasis is placed on proving a full and complete backup.

Problems in Healthcare Systems

- Scalability
- Security
- Interoperability
- Integration
- Compliance

Cloud Computing – Benefits for Healthcare

- Address scalability, security, interoperability, integration, and compliance.
- Support for rapid development and innovation, especially for mobile and IoT devices
- **Economic advantages**
  - Cloud computing provides cost flexibility in IaaS, PaaS, and SaaS, and has the potential for reduced hardware and software costs.
  - The need for additional healthcare staff with IT skills, resources, and related costs may be reduced when using cloud services.
  - Cost for skilled IT staff resources is lower in the cloud as it spreads across many customers.

Cloud Computing – Benefits for Healthcare (Cont.)

- **Operational advantages**
  - Cloud can offer better security and privacy for healthcare data and systems than on-premises systems.
    - Security controls, such as data encryption, fine-grained access controls and access logging.
  - Web access to data, avoiding the need to store information on client devices.
  - Need for scarce IT security skills within healthcare organisations has also minimised.
- **Functional advantages**
  - Offer the potential for broad interoperability and integration.
  - Are Internet-based so connecting them to other systems and applications is easy.
  - Support rapid development and innovation, especially for mobile and IoT devices.
  - Sophisticated analytic capabilities.

Healthcare Cloud Services

Healthcare Cloud Services

Cloud Services for Healthcare

- Population health management
  - Centers for Disease Control and Prevention (CDC) http://www.cdc.gov/epiinfo/cloud.html
  - McKesson http://www.mckesson.com/health-plans/population-health/
  - Cerner https://www.cerner.com/solutions/population-health-management
  - IBM Phytel Population Health Management https://www.ibm.com/watson-health/about/phytel
- Practice management support
  - Clinic to Cloud https://www.clinictocloud.com/
  - Video Demo https://www.clinictocloud.com/customer-stories
  - eClinicalWorks EHR https://www.eclinicalworks.com/products-services/the-eclinicalworks-grid-cloud/

Cloud Services for Healthcare (Cont.)

- Diagnostic support
  - EyeArt combines automated image analysis tools with a user-friendly telemedicine/cloud-based interface to address the need for a fast screening of diabetic patients for retinopathy (the leading cause of blindness). `https://www.eyenuk.com/en/products/eyeart/`
- Image handling services
  - `https://www.postdicom.com/en/services/cloud-medical`
- Medical practitioner assistance
  - IBM Watson for Oncology `https://www.ibm.com/watson-health/solutions/cancer-research-treatment`
- Patient connectivity
  - Microsoft Cloud for Healthcare `https://www.microsoft.com/en-au/industry/health/microsoft-cloud-for-healthcare`

Cloud Services for Healthcare (Cont.)

- Data distribution services
  - Snow Flake `https://www.snowflake.com/en/`
- Laboratory services
  - See VitalDx `https://www.vitalaxis.com/lab-solutions/index.html`
  - MediaLab `https://www.medialabinc.net/`
- Clinical research
  - *Medidata Clinical Cloud* is a SaaS platform solely dedicated to transforming clinical research for life science companies and patients who depend on them.
  - It currently encompasses more than eight billion clinical records from more than three million patients from more than 12,000 studies.
  - More than one million data points are added daily.
  - `https://www.mdsol.com/en/clinical-cloud`

Cloud Services for Healthcare (Cont.)

- Cloud and IoT services
  - Ericsson IoT Accelerator platform `https://www.ericsson.com/en/internet-of-things`
  - Fall detection for elderly patients
  - Health monitoring for patients with chronic disorders, such as diabetes and heart disease
  - Detection of escalating symptoms for patients with mental disorders, such as Alzheimer.
  - Devices of these types are typically connected to the Internet and can notify / alert a designated healthcare service centre with data and other pertinent incident information.
- And many more services ...